

AWS Master Data Engineer

6-Month Exhaustive Training & Certification Program

Advanced Architecture, Hands-on Labs, and MLOps

Abstract

This 24-week program is designed to transform you from a practitioner to a Master Data Engineer on Amazon Web Services (AWS). It progressively builds from core S3 data lakes to advanced AWS Lake Formation governance, integrating daily theory, weekly service-specific mini-projects, architectural use case analyses, and massive monthly Capstone projects. A tracking timetable is included at the end.

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1 Milestone 1: Core Data Infrastructure & Storage (Month 1)

Objective: Master the foundational storage options, relational databases, and NoSQL.

Week 1: Amazon S3 & Storage Foundations

- **Monday:** S3 Architecture, Storage Classes (Standard, IA, Glacier Flexible Retrieval/Deep Archive).
- **Tuesday:** Lifecycle Policies, Versioning, Object Lock, and S3 Replication.
- **Wednesday:** AWS DataSync, Storage Gateway, and AWS Snow Family.
- **Thursday: Hands-on Project:** Deploy an S3 bucket via AWS CLI, configure a 30-day lifecycle transition to Glacier, and set up event notifications to SQS.
- **Friday: Use Case & Architecture:** Design a multi-region, compliant disaster recovery storage architecture for a healthcare provider processing 10PB of X-ray images.

Week 2: Relational Databases (RDS & Aurora)

- **Monday:** Amazon RDS Architecture, Multi-AZ deployments, and Read Replicas.
- **Tuesday:** Amazon Aurora: Cloud-native storage, Global Databases, and Aurora Serverless v2.
- **Wednesday:** Database Migration Service (AWS DMS) and Schema Conversion Tool (SCT).
- **Thursday: Hands-on Project:** Deploy an Amazon Aurora Serverless v2 cluster, create a cross-region read replica, and simulate a regional failover.
- **Friday: Use Case & Architecture:** Architect a zero-downtime migration strategy from a monolithic on-premise Oracle ERP system to Amazon Aurora PostgreSQL.

Week 3: High-Scale NoSQL (Amazon DynamoDB)

- **Monday:** DynamoDB core concepts: Partitions, RCU/WCU (Provisioned vs. On-Demand).
- **Tuesday:** Data Modeling: Primary keys, Sort keys, LSI vs. GSI (Global Secondary Indexes).
- **Wednesday:** DynamoDB Streams, TTL (Time to Live), and Global Tables.
- **Thursday: Hands-on Project:** Create a DynamoDB table, design a single-table schema for a forum, write a Python Boto3 script to insert items, and configure a GSI for querying.
- **Friday: Use Case & Architecture:** Design an ingestion and serving layer for a gaming leaderboard supporting 1 million reads/writes per second globally.

Week 4: Hybrid Ingestion & Caching

- **Monday:** Amazon ElastiCache (Redis/Memcached) for database acceleration.
- **Tuesday:** DynamoDB Accelerator (DAX).
- **Wednesday:** AWS AppSync (GraphQL) and API Gateway data ingestion.
- **Thursday: Hands-on Project:** Deploy a Redis cluster via ElastiCache. Write a Python script to cache query results from RDS to reduce database load.
- **Friday: Use Case & Architecture:** Architect a high-throughput, low-latency data API for a mobile application that experiences extreme traffic spikes during live sports events.

CAPSTONE PROJECT - Milestone 1: Build an E-Commerce Multi-Tier Data Backend. Deploy Aurora Serverless for user profiles/billing, DynamoDB for active shopping carts and click-stream tracking, S3 for product images, and AWS DMS to continuously replicate Aurora data to S3 for analytics.

2 Milestone 2: Data Warehousing & Analytics (Month 2)

Objective: Deep dive into Redshift optimization, Serverless analytics, and Data Lakes.

Week 5: Amazon Redshift Architecture & Basics

- **Monday:** Redshift MPP architecture, Leader vs. Compute nodes, RA3 nodes.
- **Tuesday:** Redshift Serverless vs. Provisioned pricing and scaling.
- **Wednesday:** Data ingestion: COPY commands, Auto-Copy from S3.
- **Thursday: Hands-on Project:** Provision a Redshift Serverless endpoint. Load 10 million rows of synthetic JSON data from S3 using the COPY command with IAM roles.
- **Friday: Use Case & Architecture:** Estimate the monthly cost and recommend an architecture (RA3 vs. Serverless) for a BI workload that is heavily utilized only during end-of-month reporting.

Week 6: Redshift Optimization & Performance

- **Monday:** Distribution Styles (EVEN, KEY, ALL, AUTO).
- **Tuesday:** Sort Keys (Compound vs. Interleaved) and Compression Encodings.
- **Wednesday:** Workload Management (WLM), Concurrency Scaling, and Materialized Views.
- **Thursday: Hands-on Project:** Create three identical tables in Redshift using different Distribution Styles. Benchmark the execution time of a complex JOIN query on each table.
- **Friday: Use Case & Architecture:** Re-architect a poorly performing multi-petabyte analytics table that is causing CPU bottlenecks during complex daily dashboard aggregations.

Week 7: Data Lakehouse & Amazon Athena

- **Monday:** Amazon Athena: Serverless Presto querying directly on S3.
- **Tuesday:** AWS Glue Data Catalog: Crawlers, Databases, and Tables.
- **Wednesday:** Redshift Spectrum: Querying S3 data directly from Redshift.
- **Thursday: Hands-on Project:** Set up a Glue Crawler to catalog an S3 bucket of raw CSVs. Write an Athena query to aggregate the data, then query the same catalog using Redshift Spectrum.
- **Friday: Use Case & Architecture:** Design a cost-effective "Lakehouse" architecture for a media company wanting to store 50PB of logs in S3 while querying them alongside core business data in Redshift.

Week 8: Data Sharing & Lake Formation

- **Monday:** AWS Lake Formation: Centralized data lake governance.
- **Tuesday:** Row-level and column-level security in Lake Formation.
- **Wednesday:** Redshift Data Sharing across AWS accounts.
- **Thursday: Hands-on Project:** Configure AWS Lake Formation over an S3 bucket. Create a data permission policy that restricts a "marketing" IAM role from seeing the "PII_Email" column in Athena.
- **Friday: Use Case & Architecture:** Architect a Data Clean Room for an ad agency to merge their CRM data with third-party advertiser data without exposing raw PII to either party.

CAPSTONE PROJECT - Milestone 2: Enterprise Data Lakehouse Build. Ingest 5 years of retail data into S3 using Parquet. Catalog it with AWS Glue. Use Lake Formation to apply row/column level security policies based on department. Create Redshift Materialized Views to serve executive dashboards via Redshift Spectrum.

3 Milestone 3: Batch Processing & ETL Orchestration (Month 3)

Objective: Master Spark on EMR, Glue ETL, and Airflow orchestration.

Week 9: Amazon EMR (Managed Hadoop/Spark)

- **Monday:** EMR Architecture, Master/Core/Task nodes, and Spot Instances.
- **Tuesday:** EMR Serverless vs. EMR on EC2 vs. EMR on EKS.
- **Wednesday:** EMRFS: Using S3 as the HDFS storage layer.
- **Thursday: Hands-on Project:** Deploy an EMR cluster with Spark. Submit a PySpark job that reads data from S3, performs a distributed aggregation, and writes the output back to S3 in Parquet format.
- **Friday: Use Case & Architecture:** Design a migration strategy for a 100-node on-premise Cloudera Hadoop cluster, prioritizing cost reduction and decoupling compute from storage in AWS.

Week 10: Serverless ETL (AWS Glue)

- **Monday:** AWS Glue ETL: DynamicFrames vs. Spark DataFrames.
- **Tuesday:** Glue Studio (Visual ETL) and AWS Glue DataBrew (Data preparation).
- **Wednesday:** Bookmarks, Job tuning (DPUs), and Flex execution.
- **Thursday: Hands-on Project:** Build a visual ETL pipeline in Glue Studio that joins two tables, drops null values, transforms a timestamp, and loads the clean data into Redshift.
- **Friday: Use Case & Architecture:** Evaluate AWS Glue vs. Amazon EMR for a team of business analysts who need to build daily reporting pipelines without managing infrastructure.

Week 11: Change Data Capture (AWS DMS & EventBridge)

- **Monday:** AWS DMS architecture for continuous replication.
- **Tuesday:** Event-driven data pipelines using Amazon EventBridge.
- **Wednesday:** AWS Lambda for lightweight serverless data transformations.
- **Thursday: Hands-on Project:** Configure a DMS task with CDC enabled to stream ongoing database changes from an RDS PostgreSQL instance to an S3 Data Lake bucket.
- **Friday: Use Case & Architecture:** Architect a near-real-time synchronization pipeline from an on-premise Oracle database to Amazon Redshift for live BI reporting.

Week 12: Pipeline Orchestration (Amazon MWAA)

- **Monday:** Apache Airflow concepts: DAGs, Operators, Tasks, and XComs.
- **Tuesday:** Amazon MWAA (Managed Workflows for Apache Airflow) architecture.
- **Wednesday:** AWS Step Functions for native AWS serverless orchestration.
- **Thursday: Hands-on Project:** Write a Python DAG for MWAA that: 1. Waits for a file in S3. 2. Triggers an AWS Glue Job. 3. Runs an Athena query. 4. Sends an SNS alert on failure.
- **Friday: Use Case & Architecture:** Design a resilient, self-healing orchestration layer for a daily banking reconciliation process spanning 15 dependent systems using MWAA and Step Functions.

CAPSTONE PROJECT - Milestone 3: Automated Batch ELT Pipeline. Use EventBridge to trigger an AWS Step Function state machine upon file upload to S3. The state machine triggers a Glue DataBrew data quality check, runs an EMR Serverless Spark job to calculate complex ML features, and loads the output into Redshift.

4 Milestone 4: Streaming & Real-Time Analytics (Month 4)

Objective: Conquer unbounded data, Kinesis, Kafka, and real-time visualization.

Week 13: Message Queuing & Streaming Ingestion

- **Monday:** Decoupling with Amazon SQS and SNS (Fan-out pattern).
- **Tuesday:** Amazon Kinesis Data Streams (KDS): Shards, Partition Keys, and scaling.
- **Wednesday:** Amazon MSK (Managed Streaming for Apache Kafka) vs. Kinesis.
- **Thursday: Hands-on Project:** Create a Kinesis Data Stream. Write a Python producer script (Boto3) to stream mock IoT temperature sensor data, and a consumer to print the data.
- **Friday: Use Case & Architecture:** Architect a global telemetry ingestion point for 5 million mobile games, ensuring exactly-once processing semantics and strict message ordering.

Week 14: Real-Time Delivery (Kinesis Firehose)

- **Monday:** Amazon Kinesis Data Firehose architecture and destinations.
- **Tuesday:** Inline data transformation using AWS Lambda in Firehose.
- **Wednesday:** Format conversion (JSON to Parquet/ORC) on the fly.
- **Thursday: Hands-on Project:** Connect a Kinesis Firehose delivery stream to your KDS. Configure a Lambda function to uppercase all incoming strings, convert the data to Parquet, and output to S3.
- **Friday: Use Case & Architecture:** Design a pipeline to anonymize incoming healthcare HL7 messages in real-time before streaming them into a Redshift cluster for analysis.

Week 15: Stream Processing (Apache Flink on AWS)

- **Monday:** Managed Service for Apache Flink (formerly Kinesis Data Analytics).
- **Tuesday:** Event Time vs. Processing Time and Watermarks in Flink.
- **Wednesday:** Windowing strategies: Tumbling, Sliding, and Session windows.
- **Thursday: Hands-on Project:** Write a Flink SQL application that reads from a Kinesis Stream, groups user clicks into 5-minute Tumbling Windows, counts the clicks, and writes the output to a new stream.
- **Friday: Use Case & Architecture:** Troubleshoot a scenario where a streaming dashboard is showing inaccurate daily aggregations due to mobile devices going offline and uploading data late.

Week 16: Business Intelligence & Amazon QuickSight

- **Monday:** Amazon QuickSight Architecture and SPICE engine.
- **Tuesday:** Connecting QuickSight to Redshift, Athena, and S3.
- **Wednesday:** Row-level security (RLS) and ML Insights in QuickSight.
- **Thursday: Hands-on Project:** Connect Amazon QuickSight to your Redshift cluster. Create a real-time dashboard monitoring sales, apply a filter, and set up an ML Anomaly Detection alert.
- **Friday: Use Case & Architecture:** Design the BI access layer for an enterprise where 500 analysts need to query centralized data, ensuring regional sales managers can only see data for their specific regions (RLS).

CAPSTONE PROJECT - Milestone 4: Real-Time Fraud Detection. Build a Python script simulating credit card swipes sent to Kinesis Data Streams. Create a Managed Flink application that uses a Sliding Window to detect if a card is used in two different cities within 10 minutes, flags it, routes it via Firehose to Redshift, and displays the live fraud alerts in QuickSight.

5 Milestone 5: Machine Learning Operationalization (Month 5)

Objective: Embed AI into data pipelines and manage the MLOps lifecycle with SageMaker.

Week 17: In-Database Machine Learning (Redshift ML)

- **Monday:** Redshift ML architecture (integration with SageMaker Autopilot).
- **Tuesday:** Creating models: Regression, Classification, and K-Means.
- **Wednesday:** Model evaluation and scoring directly via SQL queries.
- **Thursday: Hands-on Project:** Use Redshift ML to train a logistic regression model on sample customer data to predict churn. Run a SQL query using the new model function to predict churn on live data.
- **Friday: Use Case & Architecture:** Architect a system to automatically forecast daily inventory needs for 1,000 retail stores using purely SQL-based pipelines to minimize data movement.

Week 18: Amazon SageMaker Foundations

- **Monday:** SageMaker Studio, Notebook Instances, and SageMaker Data Wrangler.
- **Tuesday:** Built-in Algorithms (XGBoost, BlazingText) vs. Custom Containers.
- **Wednesday:** SageMaker Endpoints: Real-time, Serverless, and Batch Transform.
- **Thursday: Hands-on Project:** Use SageMaker Data Wrangler to clean an S3 dataset. Train an XGBoost model using SageMaker Autopilot. Deploy the model to a serverless endpoint and send a test payload.

- **Friday: Use Case & Architecture:** Design a workflow for data scientists to train custom PyTorch models on GPU instances while securely accessing training data stored in a governed S3 data lake.

Week 19: MLOps & SageMaker Pipelines

- **Monday:** SageMaker Feature Store: Online vs. Offline serving.
- **Tuesday:** SageMaker Pipelines for MLOps orchestration.
- **Wednesday:** SageMaker Model Monitor (Data Drift and Concept Drift).
- **Thursday: Hands-on Project:** Build a simple SageMaker Pipeline (Preprocess → Train → Evaluate → Register). Execute the pipeline from a Jupyter notebook.
- **Friday: Use Case & Architecture:** Architect an MLOps lifecycle that detects data drift in a production recommendation engine and automatically triggers a retraining pipeline using EventBridge.

Week 20: Unstructured Data & AI Services

- **Monday:** Amazon Rekognition (Images/Video) and Amazon Textract (OCR).
- **Tuesday:** Amazon Comprehend (NLP) and Amazon Translate.
- **Wednesday:** Integrating AWS AI APIs within Lambda and Glue pipelines.
- **Thursday: Hands-on Project:** Write a Lambda function triggered by an S3 upload that sends a scanned PDF document to Amazon Textract, extracts the key-value pairs, and saves them as JSON back to S3.
- **Friday: Use Case & Architecture:** Design a scalable pipeline to process terabytes of uploaded user images, extract text (OCR), tag objects, and make the metadata searchable using OpenSearch.

CAPSTONE PROJECT - Milestone 5: End-to-End MLOps Pipeline. Use MWAA (Airflow) to orchestrate a pipeline that extracts features via Athena, pushes them to SageMaker Feature Store, trains a custom model using SageMaker Training Jobs, registers the model in the Model Registry, and sets up SageMaker Model Monitor for serving skew.

6 Milestone 6: Security, Governance & Exam Prep (Month 6)

Objective: Secure the perimeter, implement Data Mesh, and prepare for certification.

Week 21: Security, IAM & Networking

- **Monday:** AWS IAM Deep Dive: Roles, Policies, Boundaries, and SCPs.
- **Tuesday:** VPC Networking: Endpoints (Gateway vs. Interface) and PrivateLink.
- **Wednesday:** AWS KMS (Key Management Service) and AWS Secrets Manager.
- **Thursday: Hands-on Project:** Create a VPC Endpoint for S3 to ensure traffic doesn't traverse the public internet. Attempt to access the S3 bucket from an EC2 instance without an Internet Gateway.
- **Friday: Use Case & Architecture:** Design a highly secure data perimeter for a health-care provider ensuring no data exfiltration can occur to outside AWS accounts even if IAM credentials are compromised.

Week 22: Data Governance & Quality

- **Monday:** AWS Lake Formation: Cross-account data sharing and Tag-based access control.
- **Tuesday:** AWS Glue Data Quality: Automated rulesets and alerts.
- **Wednesday:** Data Lineage with Amazon SageMaker ML Lineage Tracking.
- **Thursday: Hands-on Project:** Implement an AWS Glue Data Quality ruleset within an ETL job to fail the pipeline if a required column is more than 5% null. Route failure alerts to SNS.
- **Friday: Use Case & Architecture:** Architect a Data Mesh for a global conglomerate with 5 independent domains, ensuring centralized governance via Lake Formation but decentralized data engineering.

Week 23: Compliance & Data Privacy (Amazon Macie)

- **Monday:** Amazon Macie: Automated PII discovery and classification in S3.
- **Tuesday:** Masking, Tokenization, and Crypto-based hashing in pipelines.
- **Wednesday:** AWS CloudTrail and AWS Config for auditing data access.
- **Thursday: Hands-on Project:** Enable Amazon Macie on an S3 bucket. Upload a CSV containing mock credit card numbers. Review the Macie findings report and trigger a Lambda function to quarantine the file.
- **Friday: Use Case & Architecture:** Design a pipeline that ingests third-party customer data, automatically identifies PII using Macie, tokenizes it using a KMS key in Glue, and stores the safe version in the data warehouse.

Week 24: Final Certification Preparation

- **Monday:** Review AWS Certified Data Engineer – Associate/Professional Domains.
- **Tuesday:** Whitepaper Review: Big Data Analytics Options on AWS.
- **Wednesday:** Full-length Mock Exam 1 (Identify weak domains).
- **Thursday:** Targeted review of weak domains (e.g., Kinesis vs. MSK, Redshift optimization).
- **Friday:** Full-length Mock Exam 2.

CAPSTONE PROJECT - Milestone 6: Secure, Governed Data Mesh. Build a multi-account architecture using AWS Organizations. Account A ingests streaming PII data and uses Macie/Lambda to tokenize it. Account B (Analytics) hosts Redshift. Account C (Governance) uses Lake Formation to enforce cross-account tag-based security and VPC Endpoints to lock down all traffic.

7 6-Month Progress Tracking Timetable

Week	Primary Focus	Key Project / Capstone	Status
1	Amazon S3 & Migration	S3 Lifecycle & DataSync	☐
2	Amazon RDS & Aurora	Aurora Serverless Failover Simulation	☐
3	Amazon DynamoDB	DynamoDB Single-Table Design & GSI	☐
4	Hybrid Ingestion & Caches	Cap 1: E-Commerce Multi-Tier Data	☐
5	Redshift Architecture	COPY from S3 to Redshift Serverless	☐
6	Redshift Optimization	Dist/Sort Keys Benchmark	☐
7	Athena & Lakehouse	Glue Crawler & Athena vs Spectrum	☐
8	AWS Lake Formation	Cap 2: Enterprise Data Lakehouse	☐
9	Amazon EMR	Spark Job on EMR Cluster	☐
10	AWS Glue ETL	Glue Studio Visual Pipeline	☐
11	AWS DMS & EventBridge	RDS to S3 CDC via DMS	☐
12	Amazon MWAA (Airflow)	Cap 3: Automated Batch ELT	☐
13	Amazon Kinesis Data Streams	IoT Producer/Consumer Scripting	☐
14	Kinesis Data Firehose	Firehose inline Lambda format conversion	☐
15	Managed Apache Flink	Flink Tumbling Windows on Streams	☐
16	Amazon QuickSight	Cap 4: Real-Time Fraud Telemetry	☐
17	Redshift ML	Churn Prediction SQL Models	☐
18	SageMaker Foundations	Autopilot Deployment & Endpoint Test	☐
19	SageMaker Pipelines	SageMaker Pipeline execution	☐
20	AI APIs (Textract/Rekognition)	Cap 5: End-to-End MLOps Pipeline	☐
21	IAM & VPC Endpoints	S3 Gateway Endpoint implementation	☐
22	Glue Data Quality	Automated DQ Rulesets & Alerts	☐
23	Amazon Macie	PII Discovery & Lambda Quarantine	☐
24	Exam Prep	Cap 6: Secure Data Mesh Design	☐

"Consistency is the architecture of mastery."

Best of luck on your 6-month journey to becoming a Master AWS Data Engineer!